

CS 512: Matrix Theory

Course Code	CS 512
Course Name	Matrix Theory
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MTech (CSE), MS, PhD
Mutual Exclusion	MA512, EE522
Approval	44th BoA (Nov 2021)

Course Contents

- **Background and review:** Linear system of equations, and their solutions, Linear transformation, Matrices, Determinant, Rank, Linear Vector spaces, Basis, Dimensions, Subspaces, Inner product, and orthogonality, Range space and null space, Eigenvalues and eigenvectors. Application: Examples of linear transformation such as rotation, translation, scaling, and eigen analysis. [5 Lectures]
- **Norms for vectors and matrices:** Vector norms and their properties, Matrix norms, Error analysis in linear systems, Application: Examples of neural network optimization/regularizations. [4 Lectures]
- **Eigenvalue Problems:** Condition numbers, and their application, Generalized Eigenvalue problems, Rayleigh Quotient. Application: Physical significance of eigenvalues and vectors and its relationship with PCA and Face recognition. [4 Lectures]
- **Matrix factorization and Least square problems:** Singular value decomposition, generalized pseudoinverses, QR factorization, PCA, Least square problems. Application: Examples from dimensionality reduction and Clustering. [5 Lectures]
- **Sparse matrices, their analysis, and algorithms:** Graphs and matrices, Sparse Gaussian elimination, Sparse eigenvalue, and singular value problems. Application: Relationship of sparse matrices with graph-based spectral clustering OR graph CNNs. [4 Lectures]
- **Different types and matrices, their properties, and analysis:** Symmetric, stochastic, Random Matrices, Properties of positive definite matrices, Toeplitz, and Circulant matrices. Application: Toeplitz's relationship with convolution and deconvolution networks OR DSM based graph clustering. [6 Lectures]

Text Books

1. Roger A. Horn and Charles R. Johnson, **Matrix Analysis**, Cambridge university press, 2012.
2. Gene H. Golub and Charles F. Van Loan, **Matrix computations**, 3ed Edition, John Hopkins University Press, 2012.

References

1. T. A. Davis, **Direct Methods for Sparse Linear Systems**, SIAM, 2006
2. Joel Tropp, **An Introduction to Matrix Concentration Inequalities**, 2015
3. Terence Tao, **Topics in Random Matrix Theory**, AMS, 2012
4. Lloyd N. Trefethen and David Bau III, **Numerical linear algebra**, Siam, 1997.
5. Alan J. Laub, **Matrix analysis for scientists and engineers**, Siam, 2005.
6. Harry Dym, **Linear algebra in action**, American Mathematical Soc., 2013.
7. Gilbert Strang, **Linear Algebra and its application**, 4th Edition, Cengage Learning
8. Rajendra Bhatia, **Matrix Analysis**, Springer 1997.
9. David Lewis, **Matrix Theory**, 3rd Edition, Allied Publishers, 2014.

It is a subset of the existing course EE522 with topics relevant for MTech CSE. The 2 credit structure for this course is already approved in the senate document for CSE MTech.